

Series Review from Calculus 2

here a_n denotes the entire term, not just the coefficient.

Test for Divergence:

If $\lim_{n \rightarrow \infty} a_n$ DNE or if $\lim_{n \rightarrow \infty} a_n \neq 0$ then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

P-series:

The p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges for $p > 1$ and diverges for $p \leq 1$.

Integral test:

Suppose f is a continuous, positive, and decreasing function on $[1, \infty)$. Let $a_n = f(n)$. Then

If $\int_1^{\infty} f(x) dx$ is convergent, $\sum_{n=1}^{\infty} a_n$ is convergent.

If $\int_1^{\infty} f(x) dx$ is divergent, $\sum_{n=1}^{\infty} a_n$ is divergent.

Direct Comparison test:

Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.

1. If $\sum b_n$ is convergent and $a_n \leq b_n$ for all n , then $\sum a_n$ is also convergent
2. If $\sum b_n$ is divergent and $a_n \geq b_n$ for all n , then $\sum a_n$ is also divergent

Limit Comparison test:

Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.

If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$ where c is a finite, positive number, then either both series converge or both diverge.

Alternating Series test:

If $b_n > 0$ and $\sum_{n=1}^{\infty} (-1)^n b_n$ satisfies

(1) $b_{n+1} \leq b_n$ for all n

(2) $\lim_{n \rightarrow \infty} b_n = 0$

then the series is convergent.

Ratio test:

For the series $\sum_{n=1}^{\infty} a_n$ if $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ $\left\{ \begin{array}{ll} < 1 & \text{the series is absolutely convergent} \\ = 1 & \text{the ratio test is inconclusive} \\ > 1 & \text{the series diverges} \end{array} \right.$

Root test:

For the series $\sum_{n=1}^{\infty} a_n$ if $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$ $\left\{ \begin{array}{ll} < 1 & \text{the series is absolutely convergent} \\ = 1 & \text{the root test is inconclusive} \\ > 1 & \text{the series diverges} \end{array} \right.$